Practice workbook · Run-length encoding (RLE)

Compressing a string or pixel grid with RLE, and explaining whether RLE suits given data.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-15

1. Question 1 · part b.ii · 3 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part b.ii only.

1 Computers use binary to represent and store data.

(a) Binary and hexadecimal notation can be used to represent numbers.

(i) Convert the denary number 77 to 8-bit binary.

(2)

(ii) Convert the denary number –126 to 8-bit binary using two's complement.

(2)

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(iii) Convert the binary pattern 11000110 to hexadecimal.

(2)

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1. Question 1 · part b.ii · 3 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part b.ii only.

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(b) Images are stored as binary data.

(i) A bitmap image is 400 pixels wide by 200 pixels high.

It has a colour depth of 12 bits.

Construct an expression to show how the image size, in bytes, is calculated.

Do **not** calculate the answer.

(2)

(ii) **Figure 1** is an image of an icon, set out in an 8 × 8 pixel grid.

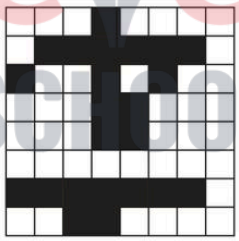


Figure 1

A run-length encoding compression algorithm uses:

- W for white pixels
- B for black pixels.

Give the result of compressing the first 4 lines of the icon using the algorithm.

(3)

Line 1 .....

Line 2 .....

Line 3 .....

Line 4 .....

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1. Question 1 · part b.ii · 3 marks

2021 NOV · Paper 01

Worked-example retry · Attempt part b.ii only.

(iii) Run-length encoding is a type of lossless compression.

State what is meant by the term **lossless** in this context.

(1)

(c) Raoul wants to record a song and store it as a digital sound file.

Figure 2 shows an overview of the process of recording a song and storing it as a digital sound file.

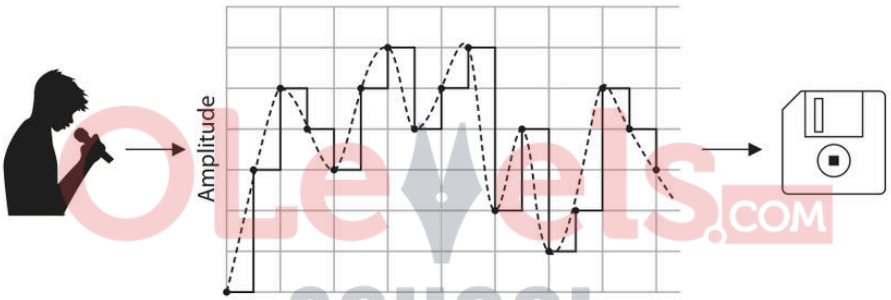


Figure 2

(i) Describe the steps taken to convert the analogue sound to a digital sound file.

(3)

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4

P 6 9 2 9 7 A 0 4 2 0

1. Question 1 · part b.ii · 3 marks · Working and answer

Working and answer

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2. Question 2 · part f.iii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part f.iii only.

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2 Many organisations use networks.

(a) One reason for networking computers is to share access to the internet.

Give **one other** reason for connecting computers in a local area network.

(1)

(b) State the purpose of network protocols.

(1)

(c) Data is transmitted in packets.

(i) Identify the layer of the TCP/IP model that would add the File Transfer Protocol (FTP).

(1)

- ☒ **A** Application
- ☒ **B** Data
- ☒ **C** Network
- ☒ **D** Transport

(ii) Give **one** task carried out by the network layer.

(1)

(d) Complete the table by adding a tick (✓) to match each item to its name.

(3)

Item	Domain Name	IP Address	URL
https://www.pearson.com			
216.33.251.195			
pearson.com			

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P 6 6 4 8 9 A 0 5 2 0

2. Question 2 · part f.iii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part f.iii only.

(e) **Figure 2** shows a network topology.

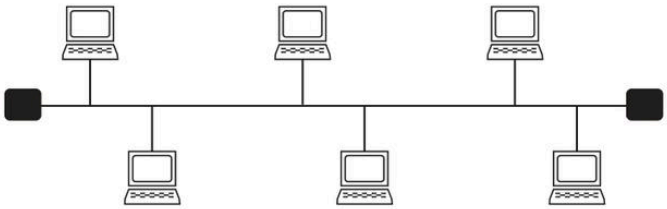


Figure 2

(i) State the name of this network topology.

(1)

(ii) State **one** benefit of using this network topology.

(1)

(iii) State **one** drawback of using this network topology.

(1)

(f) Files may be compressed using lossless or lossy algorithms.

(i) Music files are distributed over the internet.

Give **two** reasons why lossy compression is used when distributing music files over the internet.

(2)

1 .....

2 .....

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2. Question 2 · part f.iii · 1 mark

2021 JUN · Paper 01

Further practice · Attempt part f.iii only.

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(ii) Identify which **one** of these file formats uses lossy compression.

(1)

- ☒ **A** BMP
- ☒ **B** DOC
- ☒ **C** JPEG
- ☒ **D** PNG

(iii) Here is a string of data.

CCCWWWCCWWWWWWCCC

Give the result of compressing the string using a run-length encoding algorithm.

(1)

.....

.....

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(g) A video file is to be transmitted over the internet.

- The network transmission speed is 54 Mbps.
- The file size is 6 gigabytes (GB).

Construct an expression to show how the transmission time, in seconds, is calculated.

You do **not** have to do the calculation.

(4)

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(Total for Question 2 = 18 marks)

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2. Question 2 · part f.iii · 1 mark · Working and answer

Working and answer

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3. Question 3 · part e · 2 marks

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Further practice · Attempt part e only.

3 Software programs carry out tasks on computers.

(a) A headteacher uses a computer simulation to model the effect of increasing class sizes.

Explain **one** problem with using simulations to predict the effects of changes.

(2)

(b) Images are stored as files before being displayed or printed.

(i) Give **one** measurement of image resolution.

(1)

(ii) Increasing the colour depth of an image leads to an increase in the image file size.

Describe the reason for the increase in file size.

(2)

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8

P 7 2 9 3 7 A 0 8 2 4

3. Question 3 · part e · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part e only.

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(c) Software is divided into two categories.

Describe **one** difference between system software and application software.

(2)

(d) Here is an image of secondary storage.

Two files (W and Z) are stored on it.

Each file is made up of several blocks (e.g. Z1, Z2, Z3).

	W4	Z1				W2		W3	Z3		W1		Z2

Complete the image to show the state after running a defragmentation utility.

(2)

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3. Question 3 · part e · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part e only.

(e) Here is part of a file that contains electric meter readings.

04631 04984 05103 05163 05271 05383 05487 05722

Explain the effect on the file of applying a run-length encoding algorithm to this data file.

(2)

.....

.....

.....

.....

(f) A series of tasks is written in different programming languages.

Complete the table to show the correct programming language translator for each task.

(4)

Task	Translator
A guessing game that can be used on different computing platforms	
A screen driver for a new smartphone	
A new version of a spreadsheet program for sale next year	
Control software for an embedded system inside a new washing machine	

(Total for Question 3 = 15 marks)

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3. Question 3 · part e · 2 marks · Working and answer

Working and answer

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4. Question 3 · part b.iii · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part b.iii only.

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3 Computers use binary to represent and store data.

(a) Binary and hexadecimal can be used to represent numbers.

(i) Convert the binary pattern 1011 1001 to hexadecimal.

(2)

(ii) Identify the statement that is true for hexadecimal.

(1)

- ☒ **A** Hexadecimal can be processed faster by a computer than binary
- ☒ **B** Hexadecimal digits can be represented using one nibble
- ☒ **C** Hexadecimal uses less storage space in RAM than binary
- ☒ **D** Hexadecimal uses the number base 6

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(iii) Describe the binary shift that is equivalent to dividing an unsigned binary number by 4.

(2)

4. Question 3 · part b.iii · 2 marks

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Further practice · Attempt part b.iii only.

(b) **Figure 1** shows a bitmap image in a 16 × 16 grid. The image uses three colours.

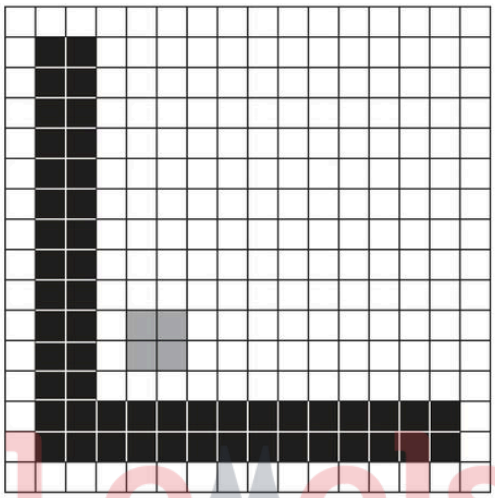


Figure 1

(i) State the smallest component of a bitmap image.

(1)

.....

.....

(ii) Construct an **expression** to show the number of images, like the one in **Figure 1**, that can be stored in 1 KB.

You do **not** need to do the calculation.

(4)

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4. Question 3 · part b.iii · 2 marks

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Further practice · Attempt part b.iii only.

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(iii) Explain why a run-length encoding algorithm is a good choice for compressing the image in **Figure 1**.

(2)

(c) The **Pigpen** cipher shown in **Figure 2** has been used to encrypt a message.

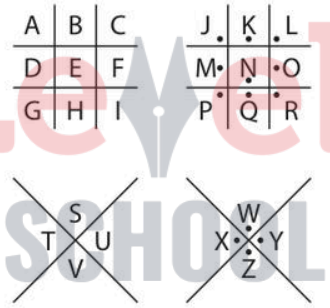
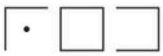


Figure 2

Complete the table to give the decrypted plain text.

(2)

Encrypted text	Plain text
	
	

(Total for Question 3 = 14 marks)

4. Question 3 · part b.iii · 2 marks · Working and answer

Working and answer

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